Executive Summary & Context: Addressing Last-Minute Cancellations at Travelers' Inn

Evaluating AI prediction pipelines to protect profit margins without expanding staff.

KEY CONTEXT & OPERATIONAL CONSTRAINTS

- Travelers' Inn faces severe margin pressure at both the City Hotel and Airport Hotel.
- Significant revenue is lost at the last minute due to fall-through bookings or no-shows.
- **Expansion of physical rooms and staff headcount is out of the question.**

CancelGuard offers a managed predictive model with a total 24-month cost of S$252,000.

// Commercial Terms & Scope

- **Solution:** Gradient-boosted ensemble providing a cancellation risk score (0–100) every morning by 7:00 AM.
- **One-Off Setup:** S$60,000 initial setup and deployment fee.
- **Monthly Subscription:** S$8,000 per month covering both hotel properties.
- **Minimum Term:** 24 months, totaling S$252,000 standard commitment.
- **Early-Bird Incentive:** 20% off the subscription for the full term if signed by 31 October 2026.

S$252,000

24-Month Total Commitment (Minimum Term)
S$60,000 Setup + S$8,000/mo Subscription (Both Hotels)

S$2.4M / YEAR

Revenue Protected with Payback in < 2 Months

20% EARLY-BIRD INCENTIVE

20% off the subscription for the full term if signed by **31 October 2026**.

Technical Assessment: CancelGuard Metrics vs. In-House Pipeline Performance

97.8% / 0.995 AUC

CancelGuard Validation Performance

VALIDATION & BENCHMARKING CONTEXT

- **Data Baseline:** Validated on a 20% hold-out sample spanning 3 years (2023–2025) of Travelers' Inn booking data.
- **Validation Objective:** We must compare these metrics directly against our custom in-house pipeline trained on the identical 2023–2025 booking records.
- **Goal:** Prove genuine incremental value before external deployment.

CONFUSION MATRIX (3,520 BOOKINGS)

	PRED. STAY	PRED. CANCEL
ACTUAL STAY	**2,141** Correct Stays	**36** False Alarms
ACTUAL CANCEL	**41** Missed Cancels	**1,302** Correct Cancels

ERROR PROFILE (77 TOTAL ERRORS)

Model evaluation shows 41 missed cancellations and 36 false alarms across the 3,520 test bookings.

The optimal model must minimize the severe S$500 relocation penalty for false alarms.

COST-BENEFIT DYNAMICS

False Alarms: Relocating a guest when a room is resold costs Travelers' Inn S$500 per instance.

Missed Cancellations: Failing to predict a cancellation costs the hotel the lost booking revenue.

S$500

RELOCATION PENALTY / FALSE ALARM

Relocation costs make false alarm minimization the critical objective for Travelers' Inn.

CANCELGUARD IMPACT (VENDOR MODEL)

36

FALSE ALARMS

In hold-out test dataset

41

MISSED CANCELLATIONS

In hold-out test dataset

Fixed Vendor Threshold Risk:

CancelGuard generated 36 false alarms and 41 missed cancellations in its hold-out test. Its fixed decision threshold cannot factor in the severe S$500 relocation penalty for false alarms.

IN-HOUSE PIPELINE ADVANTAGE

THRESHOLD TUNING CAPABILITY

Explicit Cost Matrix Calculation

Directly weights S$500 relocation penalties against lost booking revenue.

Operational Control & Cost Reduction:

Our internal pipeline allows us to explicitly calculate these costs and tune the threshold to minimize the S$500 relocation penalties, optimizing total financial outcomes.

Financial Head-to-Head: CancelGuard S$252,000 Commitment vs. In-House Total Cost of Ownership (TCO)

Weighing the external capital allocation against internal operational control.

Vendor Financial Impact

S$252,000

Total Commitment (24-Month Lock-In)

Rigid external contract with ongoing recurring overhead

- **S$252,000 total commitment** over a strict 24-month vendor lock-in contract.
- Requires ongoing subscription fees of **S$8,000 per month** regardless of seasonal volume.

In-House Financial Advantage

S$0 ADDED STAFF

Internal Resource Optimization

Leverages existing personnel and infrastructure

- Utilizes **existing internal resources and personnel** without expanding staff.
- **Eliminates external minimum term costs** and keeps technology and cost control strictly internal.
- Protects capital while allowing **customized threshold tuning** to minimize total cost of errors.

Resource Requirements & Infrastructure: Personnel, Compute, and Maintenance Needs for In-House Pipeline vs. CancelGuard

Evaluating the technical governance and operational oversight required for deployment.

CANCELGUARD // MANAGED DEPLOYMENT

Vendor-Managed Service

Externalized operational overhead with automated PMS score delivery.

INFRASTRUCTURE & GOVERNANCE PROFILE

- **PMS Score Delivery:** Fully managed service providing a daily risk score (0-100) directly into the PMS by 7:00 AM.
- **Model Maintenance:** Retrained monthly on the latest data by the vendor.

IN-HOUSE PIPELINE // INTERNAL OVERSIGHT

Full Control & Transparency

Internal ownership requiring active team maintenance and compute resource.

RESOURCE & INTERVENTION PROFILE

- **Personnel Allocation:** Requires utilizing existing internal team resources to maintain the pipeline.
- **Model Transparency:** Provides transparent feature importance and complete internal oversight of the model.
- **Custom Control:** Enables direct intervention and custom threshold adjustments without relying on external vendor support.

CancelGuard requires a six-week rollout, while an in-house solution utilizes existing resources immediately.

TI Travelers' Inn

CancelGuard Implementation Timeline

01 WEEKS 1–2
Secure connection to the booking system and historical data load.

02 WEEKS 3–4
Model calibrated on Travelers' Inn data.

03 WEEK 5
Shadow mode running alongside the current process.

04 WEEK 6
Go-live, scoring every new booking by 7:00 AM.

Operational Impact & Trade-Off

24-Month Vendor Lock-In vs. Internal Control

Relying on the vendor introduces a 6-week delay and a 24-month lock-in, whereas the internal pipeline is built and ready for immediate shadow testing.

24 MONTHS
Vendor Lock-In Contract

INTERNAL
Tech & Cost Control

STRATEGIC EVALUATION: VENDOR vs IN-HOUSE PIPELINE

Final Recommendation & Strategic Action Plan for Executive Leadership

TI Travelers' Inn

01 // FINANCIAL IMPACT

Cost Structure Evaluation

Reject the **S$252,000** CancelGuard contract to optimize capital allocation.

02 // RISK MANAGEMENT

Penalty Exposure Mitigation

Deploy the in-house pipeline tuned explicitly to minimize the **S$500 guest relocation penalty**.

03 // TECHNICAL CONTROL

Model Governance & Transparency

Maintain full internal oversight and transparent feature importance rather than relying on a managed vendor model.

04 // NEXT STEPS

Data-Driven Execution

Proceed with the internal pipeline that delivers the lowest total cost of errors based on the **2023–2025 dataset**.

EXECUTIVE DIRECTIVE // ACTION PLAN

Prioritize internal pipeline deployment to eliminate external term costs, maintain technical governance, and strictly minimize relocation penalties.

TI **Travelers' Inn**

Strategic Recommendation: Deploy In-House Pipeline over CancelGuard

Optimized decision thresholds minimize relocation penalties and total operational loss.

IN-HOUSE PIPELINE // OPTIMIZED (THRESHOLD 0.4)

S$2,751,643

Total Estimated Cost (Missed Cancellations + Relocation Penalties)

CANCELGUARD // VENDOR TOTAL COST PROFILE

S$252,000 + S$2.4M Loss

High Contract Fees (24-Mo Lock-In) + Unoptimized False Alarm Risk

KEY STRATEGIC ADVANTAGES OF IN-HOUSE DEPLOYMENT

- **Cost Minimization via Threshold Tuning:** Operating at the optimal threshold (0.4) reduces total error cost to S$2.75M, directly addressing the S$500 guest relocation penalty.
- **Avoids Vendor Lock-In & Fixed Overhead:** Rejects the S$252,000 CancelGuard contract, keeping fixed costs low while leveraging existing technical infrastructure.
- **Complete Technical & Operational Control:** Enables real-time model retraining, custom feature tuning, and full internal data governance without external dependency.

EXECUTIVE DIRECTIVE // RECOMMENDATION

Proceed with the In-House Pipeline deployment immediately. It delivers optimal financial performance (~S$2.75M cost profile) while avoiding fixed vendor commitments.

Annex: Recommended Threshold

	Threshold	Missed cancellations	False alarms	Missed cancellation loss (S$)	False alarm cost (S$)	Total cost (S$)
0	0.2	664	5193	548797.76	2596500	3145297.76
1	0.3	1286	3381	1065440.53	1690500	2755940.53
2	0.4	1963	2130	1686643.45	1065000	2751643.45
3	0.5	2725	1389	2461905.80	694500	3156405.80
4	0.6	3623	821	3361262.67	410500	3771762.67
5	0.7	4465	415	4299944.85	207500	4507444.85
6	0.8	5051	159	5012226.86	79500	5091726.86

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Recommended threshold: 0.4
Minimum estimated cost: S$ 2,751,643.45
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